

Your grade: 100%

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Next item →

1.

Suppose that $\hat{z}_k = 0.1$ and $\Sigma_{z_k} = 0.1$. What is the "normalized estimation error squared (NEES)" e_k^2 ? Enter your answer with two digits to the right of the decimal point.

1 / 1 point

0.1

Correct

Yes. This is the correct answer.
2.

What is the objective of sequential processing of measurements for multi-output systems?

1 / 1 point

☐ To improve the accuracy of the Kalman-filter state estimates when z_k is a scalar.

☐ To speed up the execution of the Kalman filter (i.e., to reduce the number of floating-point operations required) when z_k is a scalar.

☐ To improve the accuracy of the Kalman-filter state estimates when z_k is a multi-element vector.

☒ To speed up the execution of the Kalman filter (i.e., to reduce the number of floating-point operations required) when z_k is a multi-element vector.

Correct

Yes. That is correct.
3.

Which of the following is true for the results presented this week regarding using Kalman filters for prediction and smoothing?

1 / 1 point

☐ Error bounds on predicted states are the same size as those for smoothed state estimates.

☒ Error bounds on predicted states are greater than those for smoothed state estimates.

☐ Error bounds on predicted states are narrower than those for smoothed state estimates.

☐ Error bounds on predicted states may either be greater or less than those for smoothed state estimates.

Correct

Yes. This is correct.
4.

Execute the Coursera Labs Jupyter notebook for Lesson 2.3.3. Use `Ms=10` to predict ten timesteps into the future and also set `usePredict=0` to force the filter not to use future values of u when predicting the future state. What is the percentage of time that the predicted velocity error is outside of the confidence bounds? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

68.9

Correct

Yes. That is correct.
5.

Execute the Coursera Labs Jupyter notebook for Lesson 2.3.4. What is the root-mean-squared velocity error for the Kalman smoother? Enter your answer with four digits to the right of the decimal point.

1 / 1 point

0.0014

Correct

Yes. That is correct.
6.

How does the objective of the standard Kalman filter compare with the objective of a steady-state Kalman filter?

1 / 1 point

☐ The standard Kalman filter produces better state estimates than the steady-state Kalman filter and the steady-state Kalman filter uses more computation than a standard Kalman filter.

☒ The standard Kalman filter produces better state estimates than the steady-state Kalman filter and the steady-state Kalman filter uses less computation than a standard Kalman filter.

☐ The standard Kalman filter produces worse state estimates than the steady-state Kalman filter and the steady-state Kalman filter uses more computation than a standard Kalman filter.

☐ The standard Kalman filter produces better worse estimates than the steady-state Kalman filter and the steady-state Kalman filter uses less computation than a standard Kalman filter.

Correct

Yes. This is correct.
7.

Under the assumptions of the steady-state Kalman filter, which variables converge to constant values? Select all that apply.

1 / 1 point

☐ The state prediction $\hat{x}_k^-$.

☐ The state estimate $\hat{x}_k^+$.

☒ The prediction-error covariance $\Sigma_{p,k}$.

Correct

Yes. This converges to a constant matrix.

☒ The estimation-error covariance $\Sigma_{e,k}$.

Correct

Yes. This converges to a constant matrix.

☒ The Kalman gain vector L_k .

Correct

Yes. This converges to a constant matrix.
8.

Execute the Coursera Labs Jupyter notebook for Lesson 2.3.5. What is the root-mean-squared estimation error of the velocity state using the steady-state Kalman filter? Enter your answer with four digits to the right of the decimal point.

1 / 1 point

0.0024

Correct

Yes. This is correct.
9.

Suppose that a **continuous-time system** has steady-state solution $\Sigma_x = 0.1$, $C = 1$, and $S_v = 0.1$. What is the steady-state Kalman gain L ? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

1

Correct

Yes. That is correct.
10.

Suppose that a **continuous-time system** has $A = -0.1$, $C = 1$, and steady-state Kalman gain $L = 0.1$. What is the pole location a of the steady-state Kalman filter? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

-0.2

Correct

Yes. That is correct.